

Self-Supervised Approach for Few-shot Hand Gesture Recognition

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ABSTRACT

Data-driven machine learning approaches have become increasingly used in human-computer interaction (HCI) tasks. However, compared with traditional machine learning tasks, for which large datasets are available and maintained, each HCI project needs to collect new datasets because HCI systems usually propose new sensing or use cases. Such datasets tend to be lacking in amount and lead to low performance or place a burden on participants in user studies. In this paper, taking hand gesture recognition using wrist-worn devices as a typical HCI task, I propose a self-supervised approach that achieves high performance with little burden on the user. The experimental results showed that hand gesture recognition was achieved with a very small number of labeled training samples (five samples with 95% accuracy for 5 gestures and 10 samples with 95% accuracy for 10 gestures). The results support the story that when the user wants to design 5 new gestures, he/she can activate the feature in less than 2 minutes. I discuss the potential of this self-supervised framework for the HCI community.

CCS CONCEPTS

• **Human-centered computing** → **Gestural input**; *Ubiquitous and mobile devices*; • **Computing methodologies** → *Machine learning*.

KEYWORDS

gesture recognition, wearable devices, self-supervised learning

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1 INTRODUCTION

Because of the availability of machine learning toolkits with easy-to-understand interfaces, data-driven machine learning methods are being integrated into human-computer interaction (HCI) systems, classified as a simple first method. However, unlike traditional machine learning tasks (e.g., image classification [3], and speech

recognition [6]), HCI research has proposed novel interactions and a use case of sensors, which requires the collection of new datasets for each study. The newly collected datasets tend to be small in amount. Therefore, it results in low performance or places a heavy burden on participants of user studies to collect labeled data per user.

Traditional machine learning communities focus on the self-supervised learning (SSL) approach [2, 4, 8] instead of conventional supervised methods, which require humans to spend large amounts of time collecting labeled data. SSL involves two steps. The first step is pretraining, in which unsupervised representation learning is performed on unlabeled data, which are easy to collect in large quantities. The second step involves fine-tuning, in which supervised learning is performed on a small amount of labeled data the user needs to prepare his/her time to collect. In speech recognition, the SSL method, which uses only 10 minutes of labeled data, is comparable to the supervised method, which uses 100 hours of labeled data[1]. The contribution of SSL has been justified as effective in a few language recognition tasks when it is challenging to prepare a large labeled dataset. However, in this study, I hypothesized that it would also be effective in the machine learning tasks performed in HCI research, where collecting a large labeled dataset is difficult.

In this study, I demonstrated the effectiveness of the self-supervised approach by taking hand gesture recognition with wrist-worn devices as the first representative HCI task. Based on a small sample size, the proposed system achieved over 95% accuracy (five samples with 95% accuracy for five gestures and ten samples with 95% accuracy for ten gestures) in recognizing fine-grained gestures. Furthermore, the results support the story that when the user wants to use five gestures, he/she can activate the feature in less than 2 minutes (assuming it takes 3 seconds per sample collection and five gestures are collected). Finally, I discuss the suitability of a self-supervised framework for other HCI tasks.

2 SYSTEM

2.1 Overview

Figure 1 shows the entire process, including the pretraining step using unlabeled data, the fine-tuning step with labeled data, and gesture recognition. Feature extractors were acquired by pretraining (i.e., unsupervised representation learning). First, the user collected a large amount of unlabeled data. However, this step did not require the user's time because the latter went about his/her everyday life while wearing the smartwatch. Unsupervised representational learning utilizes unlabeled data collected during the user's everyday life to make a feature extractor.

Since labeled data collection requires the user's attention, this step is an actual burden for the user to use the system. Labeled

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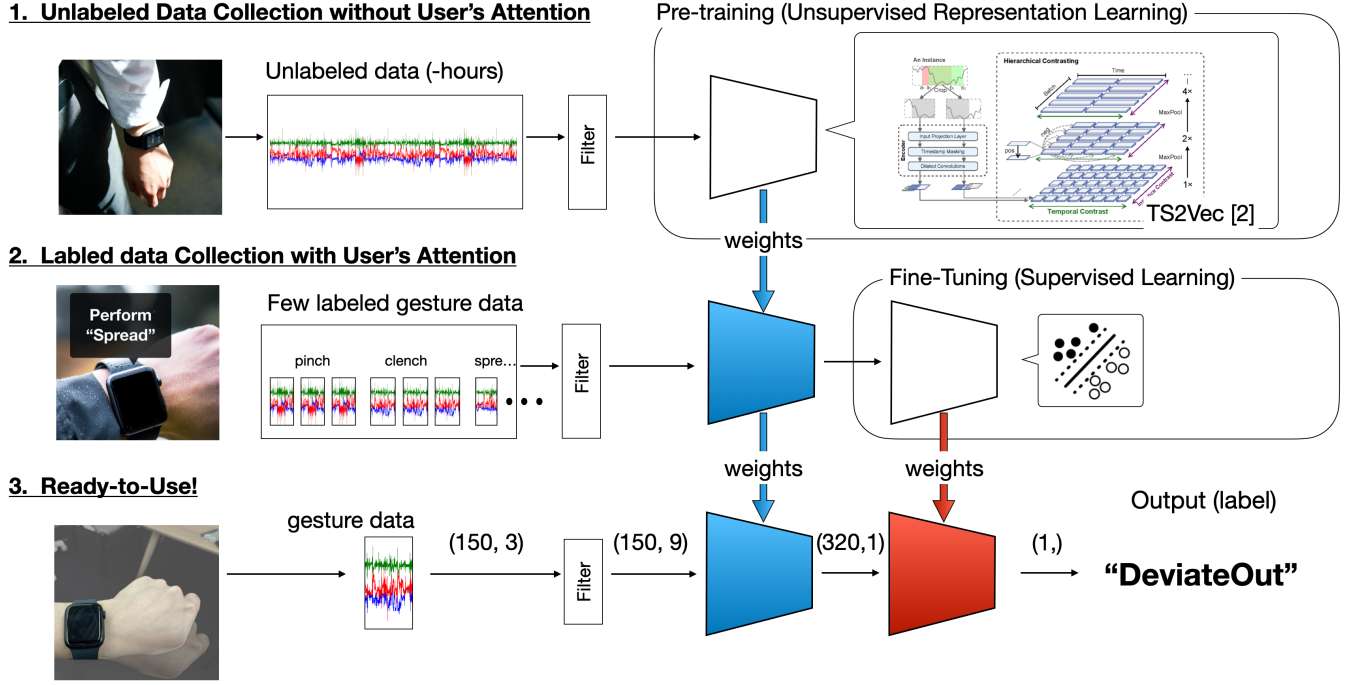


Figure 1: This figure shows the flow of the system’s data processing and the steps involved in making the gesture recognition system available to the user. The first step of data collection for pretraining is completed by simply putting on the smartwatch and living with it. The second step requires collecting a small number of labeled training samples. The user needs to prepare a few minutes to collect. The pretraining image is from TS2Vec paper [2].

training data were converted to feature vectors (i.e., representations) by the pretrained feature extractor and used to train a simple classifier (support vector machine).

Finally, the user utilized the feature extractors and classifiers created in steps 1 and 2 to perform gestures of his/her own design. The actual burden experienced by the user in this process is that collecting a small amount of labeled data takes only a few minutes (for five different gestures, 25 times 5×5 , or about 2 minutes).

2.2 Pretraining using Unlabeled Data

2.2.1 Unintended Data Collection in Daily Life. An Apple Watch from Apple inc. was used in the experiment, and the data were collected from a 3-axis accelerometer at 100 Hz using Sensorlog [9]. The user went about his/her everyday life while the data collection mode was activated. A threshold was set on the amplitude magnitude to remove the data when the user was at rest, such as using a personal computer or sleeping. As a result, the 24 hours of collected data were reduced to 120 minutes. Considering that a gesture lasts less than a second, I set the length of one sample to 150 frames, so the 120-minute data was divided into 4886 data with 150 frames and used for training (the data shape at this point was $(4886, 150, 3)$).

Following previous work that attempted few-shot gesture recognition using an Apple Watch, each channel was filtered using three Butterworth bandpass filters (22–8 Hz, 8–32 Hz, and 32 Hz), which

changed the data shape from (150×3) to (150×9) . The same filter was used for labeled data as well.

2.2.2 Unsupervised Representation Learning. TS2Vec [2] was used for learning representations without labels, which supported multivariate time series data and learned representations of time series data. TS2Vec outputs timestamp-level representation (number of frames $\times 320$ dim), but the gesture classification task required instance-level representations obtained by max pooling over all timestamps. Therefore, I obtained instance-level representations $(320, 1)$ by max pooling over all timestamps. Learning for 120 minutes of data converged in about 30 seconds.

2.3 Fine-tuning on Labeled Data

Fine-grained hand gestures are defined as finer-grained and more complex than trajectory-based gestures (e.g., line, square, circle, and star). Compared to the trajectory-based gestures, they are considered difficult to recognize using dynamic time warping (DTW) and hidden Markov models (HMM). In this paper, I use ten fine-grained gestures introduced by Xu et al. [10], consisting of Pinch, Clench, Finger-snap, Spread, Pinky-pinch, RotateOut, RotateIn, DeviateOut, DeviateIn, and Extend. I did not use Double-Tap or Double-Clench, which could be represented by combinations of smaller unit gestures this time.

For the experiment, 25 samples were collected per fine-grained hand gestures. Each sample lasted approximately 3 seconds, including margins before and after the gesture. Simple peak detection was performed to detect when the gesture occurred, and 150 frames were cut from a sample of about 3 seconds so that it was centered (150 x 3). The filter used in pretraining results in 9 channels, which are transformed with TS2Vec to become instance-level features consisting of 320 dimensions. The support vector machine included in Scikit-learn [7] was used to classify those samples processed by TS2Vec.

3 EXPERIMENTS

3.1 T-SNE Visualization

Figure 2 shows a T-SNE visualization of the collected labeled samples. Although T-SNE is not given labels, it successfully separates different gestures in just two planes. Therefore, I expected that supervised learning, which can use labels, would be able to classify them.

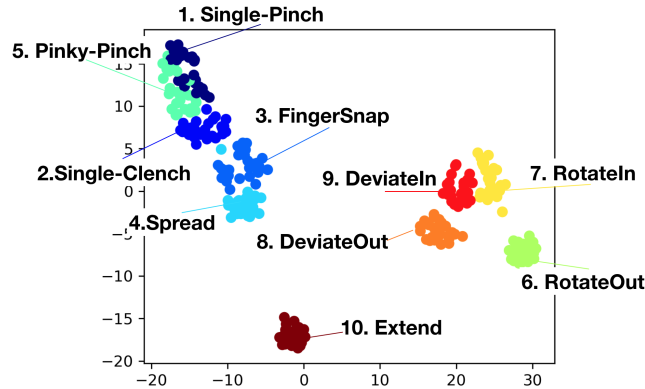


Figure 2: This figure shows labeled data for training, transformed by a pre-trained feature extractor and mapped into a two-dimensional space using t-SNE. No labels are given in the t-SNE dimensionality reduction process, but each gesture is mapped in an easily separable manner. The gestures are mapped in an easily separable manner. Similar gestures are mapped nearby.

3.2 Classification

Figure 3 shows the evolution of accuracy with respect to the number of training samples. The figure contains ten plots, each with a different number of gestures to be recognized, from 2 to all 10. When a limited number of gestures are used (2-9), the accuracy for all combinations of gestures is calculated by cross-validation and averaged. All data except the training sample were used for the test (e.g., when the number of training samples was 2, the test data were the remaining 23). First, in all conditions (number of gestures), the two training samples achieved an accuracy of more than 85%. Experimental results also show that when fewer than five gestures are recognized, as long as there are five training samples, recognition is achieved with more than 95% accuracy.

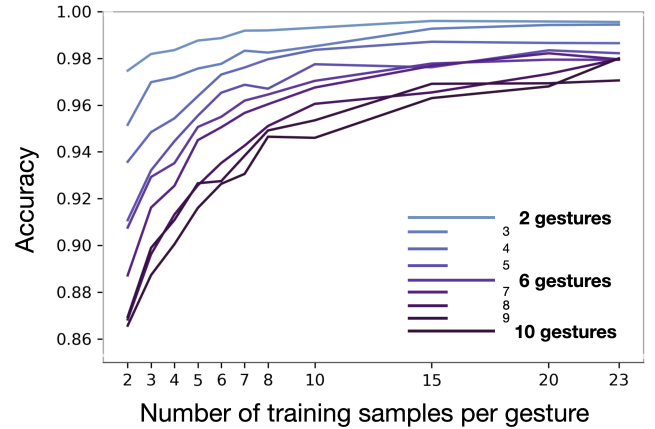


Figure 3: The number of samples to obtain sufficient accuracy depends on the number of gestures to be recognized. 95% accuracy for 5 gestures needs 5 samples per gesture. 95% accuracy for 10 gestures 10 samples.

The results support the story that when the user wants to use five gestures, he/she can activate the feature in less than 2 minutes (assuming it takes 3 seconds per sample collection and five gestures are collected). Although recognizing a set of 10-sample fine-grained gestures is considered a difficult task, the results show that with 10 training samples, this system can recognize them with more than 95% accuracy.

4 DISCUSSION AND CONCLUSION

I have proposed a self-supervised approach using a specific HCI task as an example. The user's actual data collection experience only lasted a few minutes, and multiple gestures were recognized with high accuracy. This framework can potentially exploit machine learning performance for various tasks in HCI while minimizing the burden on the user. Repeating the same specified actions with a new sensor developed for the user can be physically and mentally demanding. However, if the sensor is wearable, such as a smartwatch, it may not be a significant burden to live with it on. Myoelectric sensors are generally small and wearable, and electroencephalograph (EEG) sensors are a bit depressing to wear around the city but not too much of a burden to wear while working at a desk. For a task with a well-developed dataset (human-activity detection), Haresamudram et al. used contrastive predictive coding [5]. This framework, which consists of pretraining using a large amount of unlabeled data and fine-tuning using a small amount of labeled data, is likely to become mainstream in the HCI field.

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REFERENCES

- [1] Alexei Baevski, Henry Zhou, Abdelrahman Mohamed, and Michael Auli. 2020. wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations. *CoRR abs/2006.11477* (2020). arXiv:2006.11477 <https://arxiv.org/abs/2006.11477>

- [2] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. 2020. A Simple Framework for Contrastive Learning of Visual Representations. *arXiv preprint arXiv:2002.05709* (2020).
- [3] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. 2009. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*. Ieee, 248–255.
- [4] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *ArXiv abs/1810.04805* (2019).
- [5] Harish Haresamudram, Irfan Essa, and Thomas Ploetz. 2021. Contrastive Predictive Coding for Human Activity Recognition. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 5 (2021), 1 – 26.
- [6] Vassil Panayotov, Guoguo Chen, Daniel Povey, and Sanjeev Khudanpur. 2015. Librispeech: An ASR corpus based on public domain audio books. In *2015 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 5206–5210. <https://doi.org/10.1109/ICASSP.2015.7178964>
- [7] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. 2011. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research* 12 (2011), 2825–2830.
- [8] Steffen Schneider, Alexei Baevski, Ronan Collobert, and Michael Auli. 2019. wav2vec: Unsupervised Pre-training for Speech Recognition. In *INTERSPEECH*.
- [9] Bernd Thomas. 2022. SensorLog. <http://sensorlog.berndthomas.net/>. (Accessed on 07/21/2022).
- [10] Xuhai Xu, Jun Gong, Carolina Brum, Lilian Liang, Bongsoo Suh, Shivam Kumar Gupta, Yash Agarwal, Laurence Lindsey, Runchang Kang, Behrooz Shahsavari, Tu Nguyen, Heriberto Nieto, Scott E Hudson, Charlie Maalouf, Jax Seyed Mousavi, and Gierad Laput. 2022. Enabling Hand Gesture Customization on Wrist-Worn Devices. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (*CHI '22*). Association for Computing Machinery, New York, NY, USA, Article 496, 19 pages. <https://doi.org/10.1145/3491102.3501904>